



Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Connect the SA market story to evidence and next actions

Situation - Complication - Resolution | evidence and gaps

01 Situation | establish the market baseline

Have: demand history, regions and operators (pp.3-4,12,14).

Gap: recent fuel balance, finished imports and entry ports.

Close: matched-year balance and exact data reconciliation. W1-W2.

02 Complication | quantify what could change

Have: provisional vehicle, power-diesel and supply calculations.

Gap: calibrated driver reversals, refinery cases and route costs.

Close: define levers in W1; test their litres/import effects in W2-W4.

03 Resolution | establish Vopak opportunity

Have: regional competitor footprint and published tank stocks.

Gap: actual share, customer access and additional contestable volume.

Close: client flows + cost/access tests; investment follows the flow case.

Evidence and implications

01 | Situation

Establish demand, domestic output and finished imports; locate the economically reachable market.

02 | Complication

Test demand-driver reversals, refinery cases and the relative cost and feasibility of competing routes.

03 | Resolution

Establish actual Vopak share and additional contestable customer flows. Investment follows the flow case.

Next steps | proposed owners

Manish: close baseline/source flags and define levers.

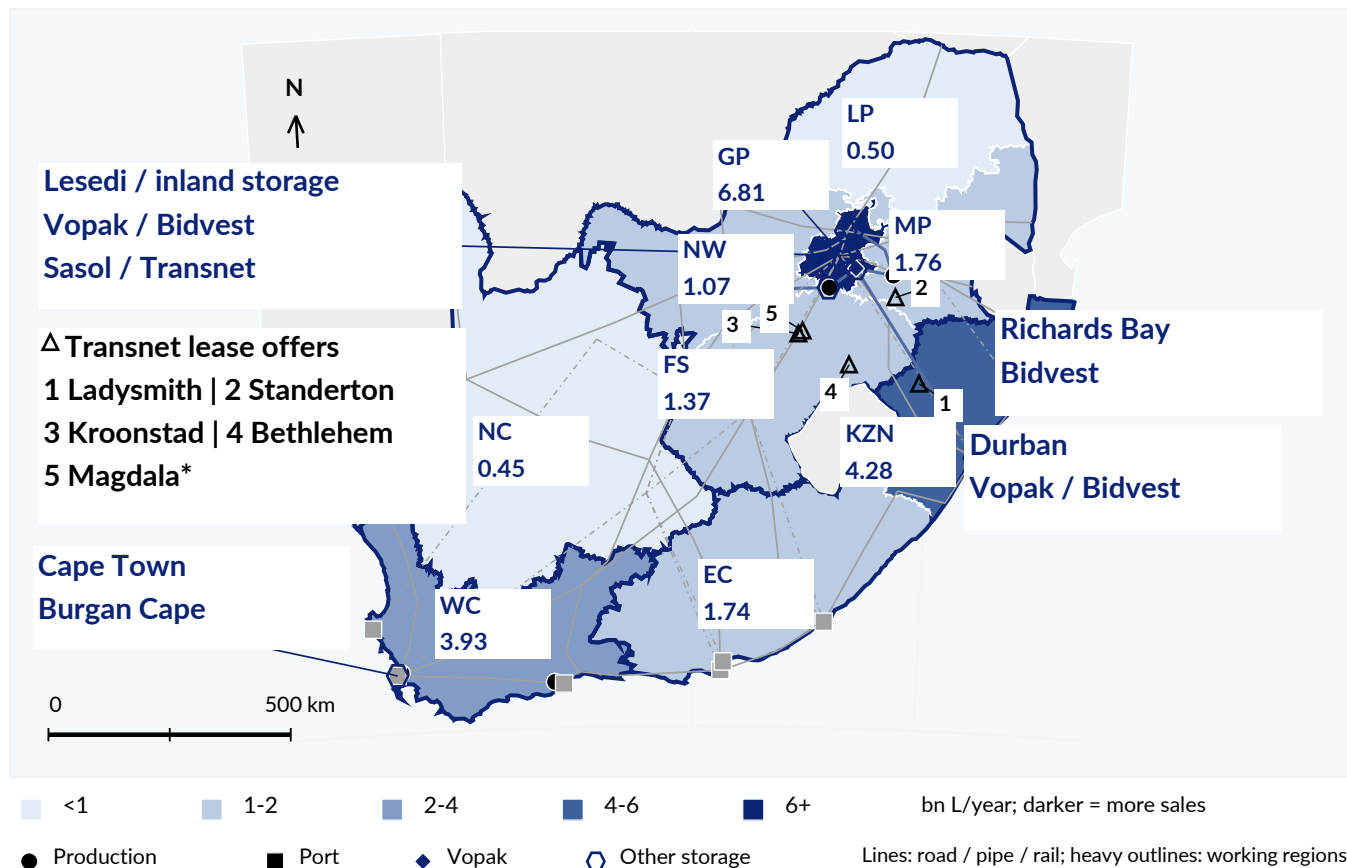
Nigel: obtain client flows and choose the first corridor.

Week windows remain proposed.

Week windows are targets. Partner propositions require testing; they are not established findings.

Locate fuel demand alongside the infrastructure serving it

Petrol + diesel sales by province | 2022, bn litres/year



Evidence and implications

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Regions are working groups

Province boundaries define reporting groups; they do not establish terminal catchments.

03 | Sales are historical demand evidence

The map is sourced provincial sales, rather than a current-year outlook or commercial accessibility result.

Next steps | proposed owners

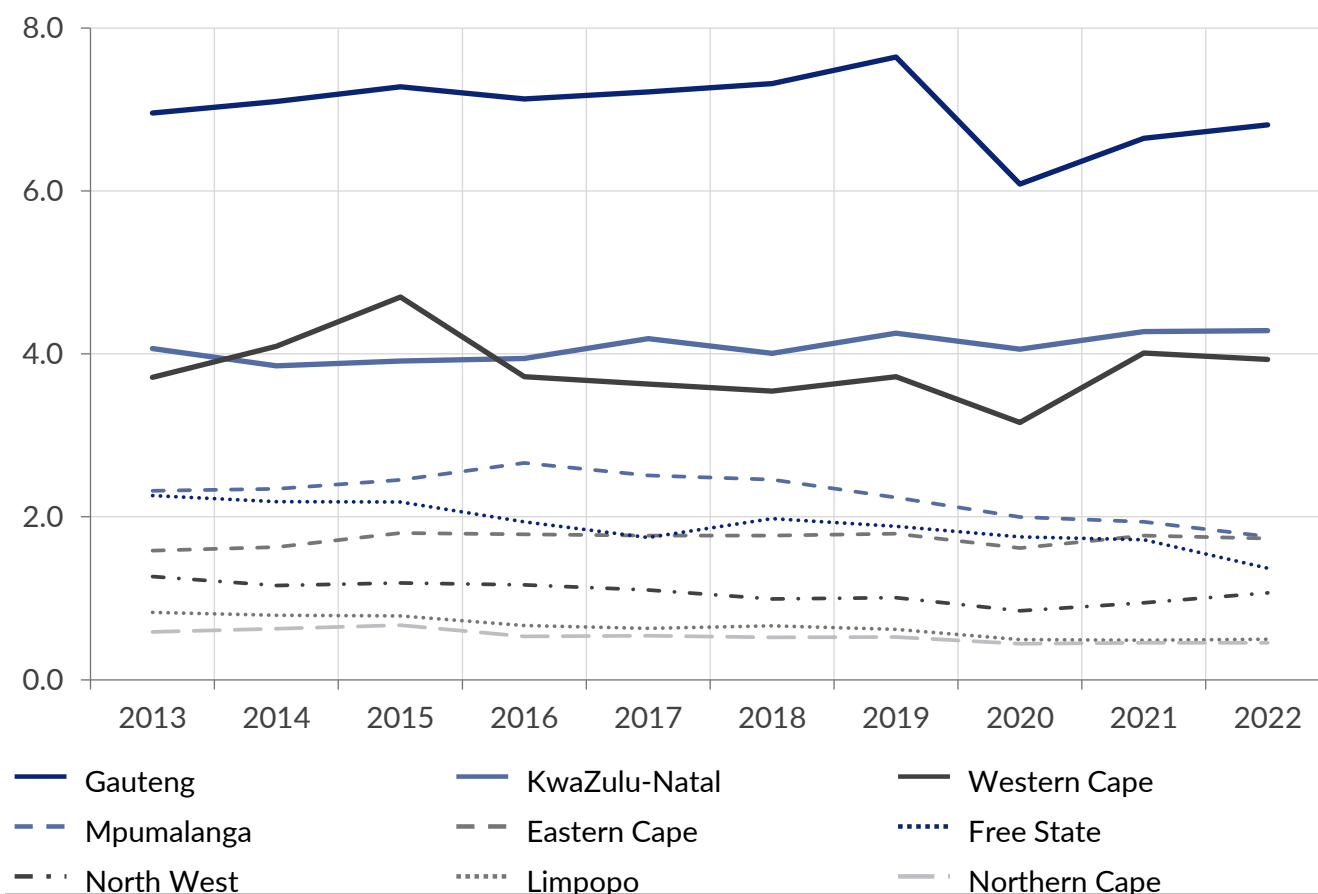
Manish: source a later complete year and test actual destinations, route costs and customer access.

Lease offers are not operating capacity. *Magdala RFP location narrative conflicts; briefing coordinate used. Jet excluded.

Source: DMPR 2022 sales; geoBoundaries/OCHA/MDB 2020 provinces (CC BY 3.0 IGO); operator sources on slide 12.

Track petrol and diesel sales by province over time

Annual provincial sales | 2013–2022, bn litres/year



Evidence and implications

01 | History ends in 2022

The latest complete provincial year in this extract is 2022.

02 | 2023 is incomplete

Only Q1 is present; it is excluded from annual comparisons.

03 | Six years are flagged

Provincial sums differ from national sales in 2013, 2014, 2015, 2017, 2018 and 2021.

Next steps | proposed owners

Manish: reconcile each flagged value to its original workbook cell and log old value, proposed value and reason.

Reconcile domestic supply and finished imports

Situation | national balance and import entry ports

EVIDENCE GAP | analysis specification, not a populated result

Domestic production versus finished imports

Matched recent demand, output, imports, exports and stocks; actual provincial origin unavailable.

Output: $\text{Production} + \text{imports} - \text{exports} - \text{stock build} = \text{consumption by fuel/year}$; unit/coverage bridge and residual explained.

Import entry ports and origin

Product-specific finished imports by port/year and source country; liquid bulk is not all fuel. Output: National finished imports tied to evidenced port allocations; unallocated remainder explicit.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Energy balance 2007-2021 staged; provisional supply engine; p6 provincial source illustration.

03 | What this page will resolve

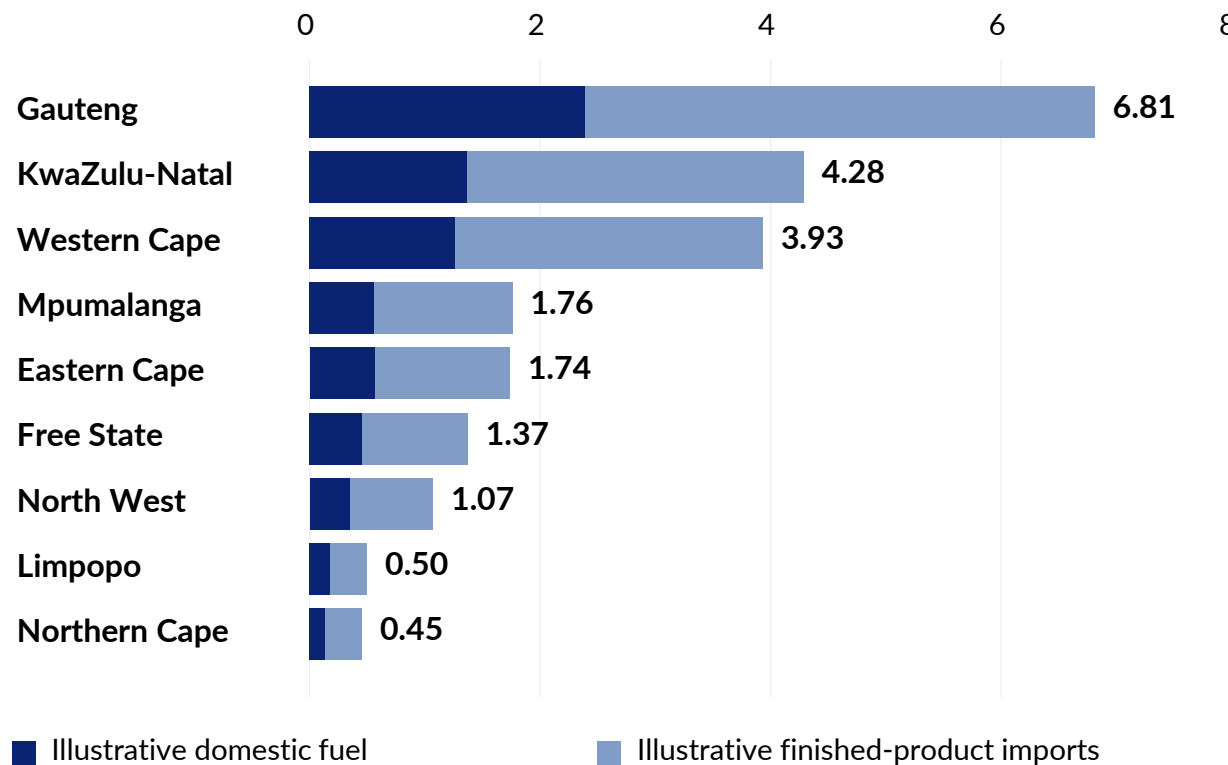
A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Manish: reconcile the national fuel balance, then bridge finished imports to entry ports. Nigel: secure source access. Explain all unit conversions and unallocated volumes.

Shape the provincial split between domestic fuel and imports

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

Evidence and implications

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

Next steps | proposed owners

Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

Translate demand drivers into explicit fuel levers

Complication | power, freight, fleet and economic activity

EVIDENCE GAP | analysis specification, not a populated result

Power-generation diesel and potential reversal

Historical GWh-to-litres tie; baseline/alternative dispatch, units, dates and incremental demand effect.

Road-to-rail freight shift

Specific intervention with baseline/alternative mode shares, dates and diesel-intensity bridge; no duplicate mileage reduction.

BEV, hybrid, efficiency and charging effects

Opening-fleet/cohort tie; drivetrain-specific L/100km; dated uptake alternatives and litres effect.

GDP, fuel prices and kilometres travelled

Explicit historical/forecast boundary and consuming function; calibrated sensitivity without duplicate activity reductions.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Eskom OCGT observations 2022-2026 staged; engine capacity/load factor/efficiency path.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Manish: define baseline, alternatives, units, dates and consuming equation for each lever. Nigel reviews; calibrate against source observations.

Test refinery availability and plant-specific scenarios

Complication | refinery closures, restart proposals and Sasol / Natref

EVIDENCE GAP | analysis specification, not a populated result

Refinery closures, availability and SAPREF proposal

Current output/yields/status; dated closures/restarts; SAPREF project evidence; 558 extra refinery rows unresolved.

Output: Unique country/year/scenario keys with source cells; actual/proposed distinction; per-refinery output tie and import sensitivity.

Secunda gas/MRG and Natref future

MRG timing and causal liquid-fuel effect; Natref future not established; partner propositions untested.

Output: Documented plant-level mechanism and dated scenario; no gas-to-liquid displacement inferred without evidence.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Refinery assumptions and utilisation 2008-2050; provisional supply engine; duplicate-row diagnostics.

03 | What this page will resolve

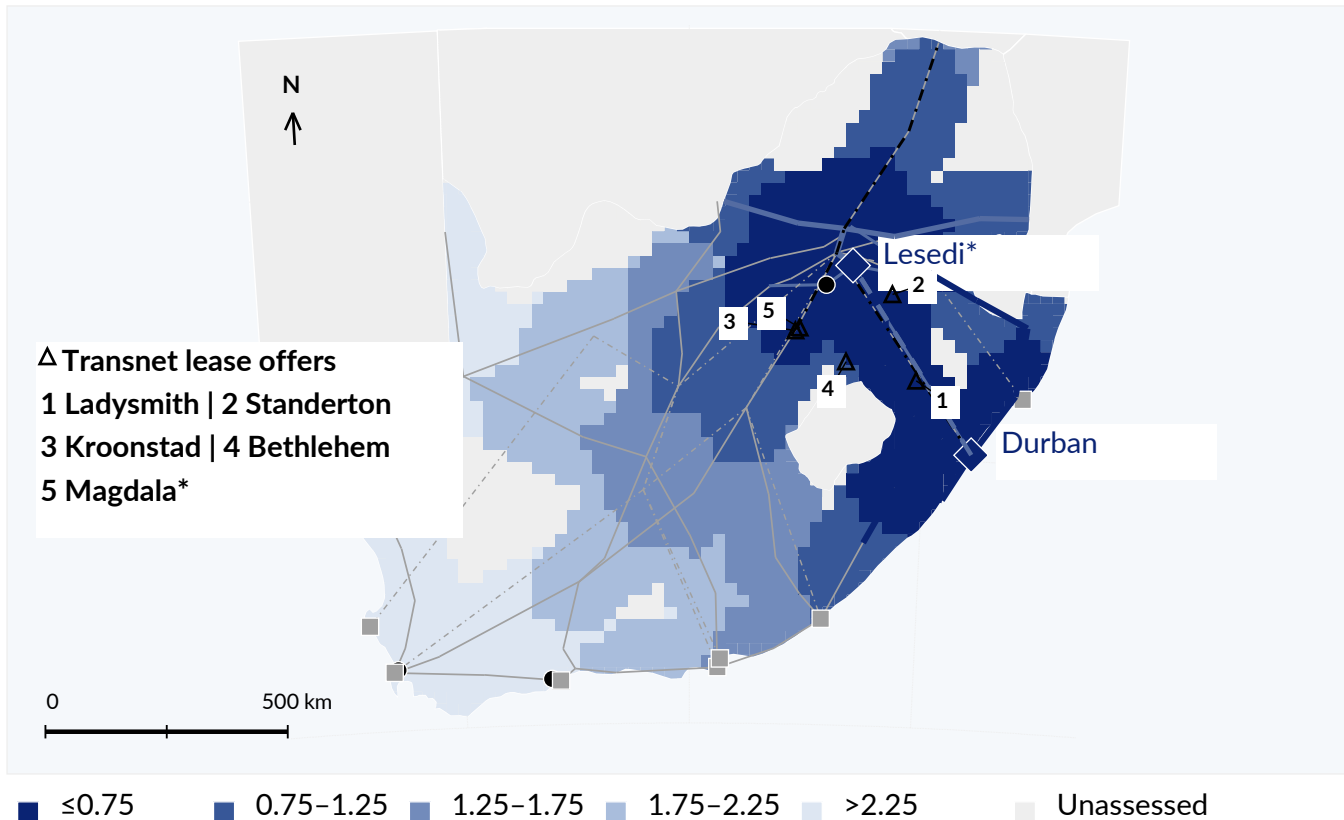
A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Manish: resolve duplicate refinery keys and source plant status. Henry: review the gas/liquid mechanism. Nigel: agree conditional refinery scenarios.

Map conditional market access from Durban and Lesedi

Road-delivery accessibility | illustrative transport cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

Evidence and implications

01 | Darker cells mean lower example cost

The surface shows illustrative delivered road-transport cost in R/litre.

02 | Other modes remain context

Pipeline and rail costs, capacity and access are not evaluated in this surface.

03 | Physical proximity is insufficient

No operational or commercial catchment is validated by this illustration.

Next steps | proposed owners

Manish and Nigel: select the first destination and obtain dated route tariffs, quotations and access evidence.

Compare routes on the same delivered-cost basis

Complication | Durban-NMPP-Lesedi, road / rail, Matola and Walvis Bay

EVIDENCE GAP | analysis specification, not a populated result

Durban-NMPP-Lesedi versus road/rail, Matola and Walvis Bay

Comparable route costs, NMPP capacity/tariff, actual rail service and cross-border feasibility.
Output: Comparable delivered R/litre with cost components, capacity/access limits and dated sources; no national optimisation claim.

Evidence required before quantification

Schematic appendix routes and illustrative Durban/Lesedi road-cost map.
No operational route allocation or delivered-cost engine.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Schematic appendix routes and illustrative Durban/Lesedi road-cost map.

03 | What this page will resolve

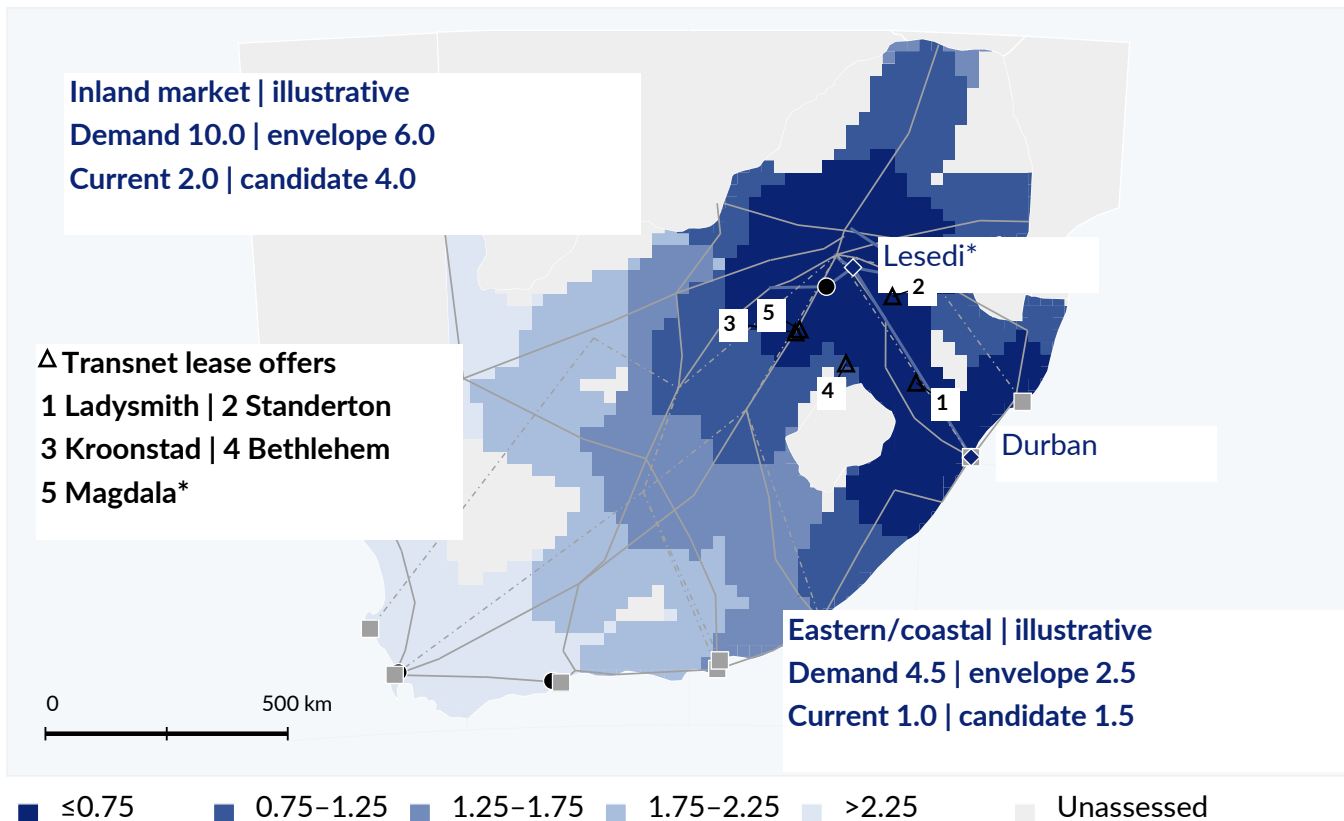
A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Nigel: choose the first product and customer destination. Manish: compare dated route costs, capacity and access on the same R/litre basis. Henry reviews.

Separate demand, accessible volume and Vopak flows

Illustrative road cost (R/litre) and market volumes



Receipts 2.8 + 2.0 – transfer 1.8 = unique demand 3.0

Example rates: R0.15/L + R0.002/L/km. Grey = unassessed. Volumes: bn L/year.

Source: Illustrative CSVs and road transport rates; no actual Vopak throughput or quoted transport costs supplied.

Evidence and implications

01 | Balance arithmetic is illustrative

Example demand 21.0 less domestic production 7.0 gives imports 14.0, before exports and stocks.

02 | Transfers count once

Example Durban receipts 2.8 plus Lesedi receipts 2.0 less shared transfer 1.8 gives unique served demand 3.0.

03 | Candidate volume is conditional

The example envelope is 8.5 and additional candidate volume 5.5; neither is forecast capture.

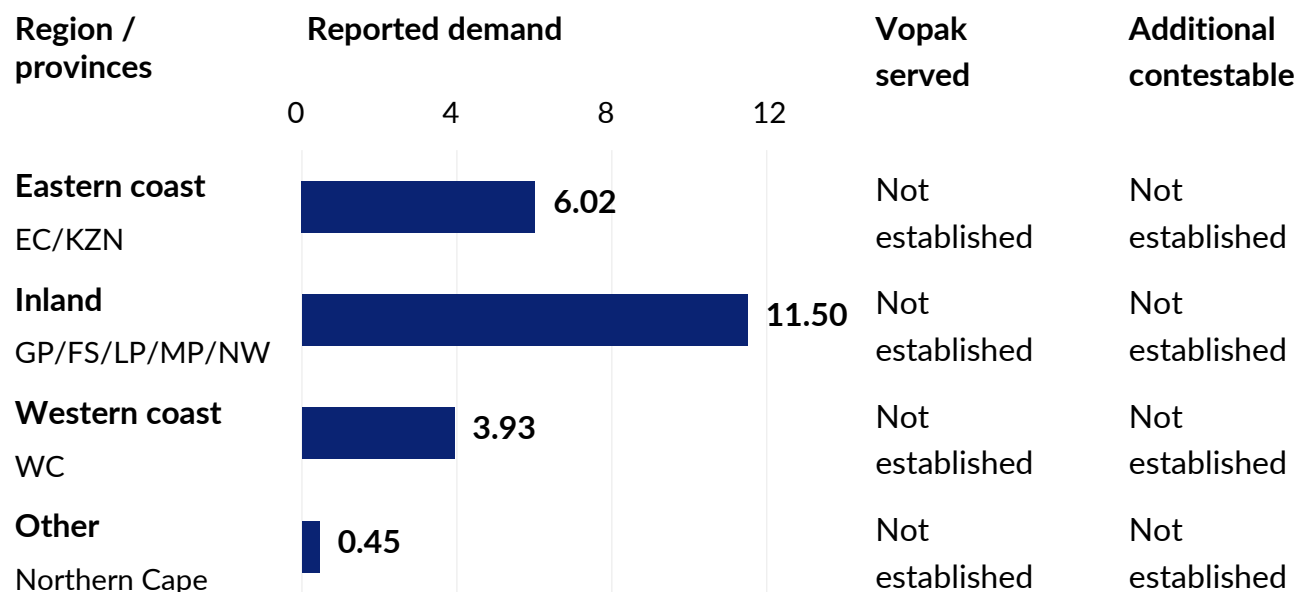
Next steps | proposed owners

Nigel: request client flows. Manish: replace examples with matched-year product and destination records.

Size each regional market and locate competing storage

All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded



To calculate share: unique Vopak customer deliveries / same-year regional demand.

Contestable demand needs route cost, compatible tanks, contracts and switching evidence.
Count Durban-Lesedi transfers once. Region boundaries are working groupings, not catchments.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.
Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].
Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

Establish actual share and additional contestable demand

Resolution | unique deliveries, reachable customers and commercial access

EVIDENCE GAP | analysis specification, not a populated result

Vopak market share at Durban and Lesedi

Unique customer deliveries, receipts, shared transfers, destinations and matched period.

Output: Separate terminal throughput and unique served-demand shares; matched-year/product denominator; traceable records, no capacity-as-share.

Additional commercially contestable market

Usable product-compatible tanks, spare routes, contracts, switching, customer overlap and Northern Cape inventory.

Output: Accessible demand by product/region/period; explicit assumptions and unassessed residual; competitors volume is not automatically capturable.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Sourced regional sales p12; published capacity p14; illustrative transfer arithmetic p11.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

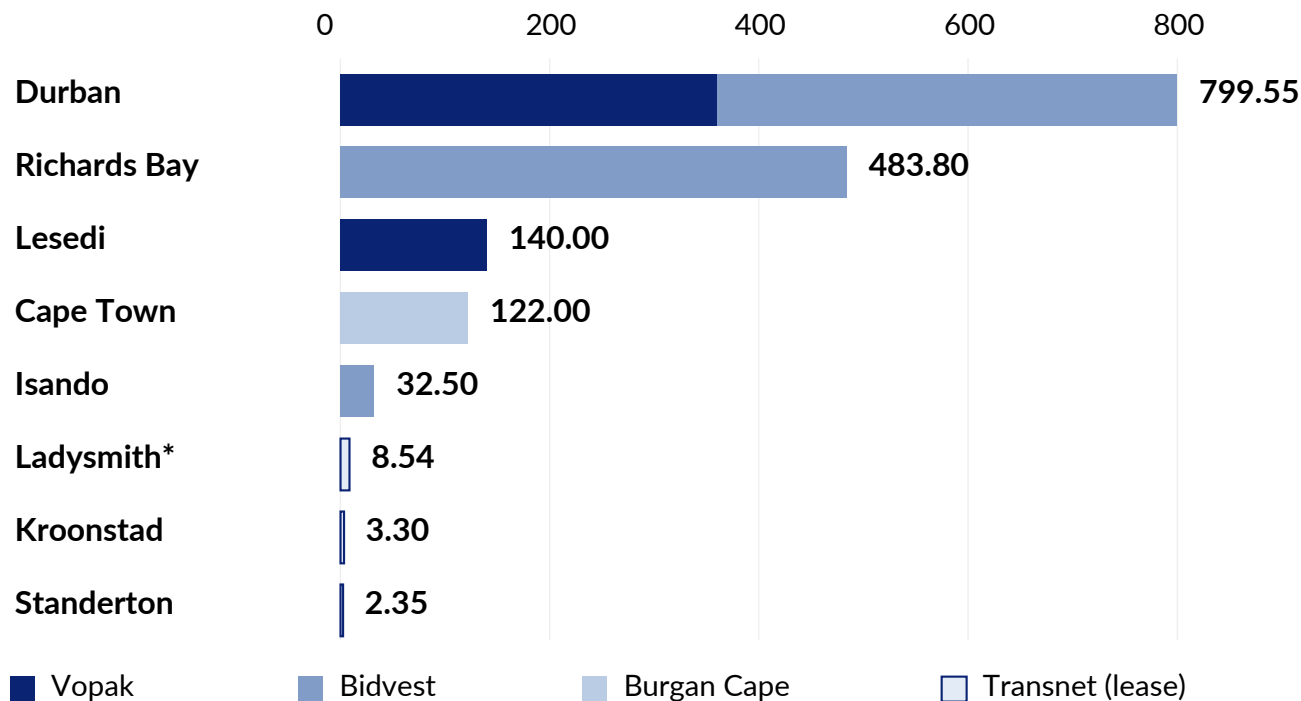
Next steps | proposed owners

Nigel: request client flow and destination records.
Manish: reconcile transfers, calculate unique served demand, then test additional customer access and competing storage.

Compare published storage capacity by location

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Borgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

Evidence and implications

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

Next steps | proposed owners

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

Gate investment on evidenced incremental customer flows

Resolution | optional service and investment assessment

EVIDENCE GAP | analysis specification, not a populated result

New storage and investment scope

Incremental accessible flows, turnover/peak inventory, product compatibility, cost and commercial return criteria.

Output: Throughput-to-usable-stock bridge with seasonality, inventory days and dispatch limits; downside sensitivity; no investment claim from import deficit alone.

Evidence required before quantification

Partial gross/lease stocks, without client handling/inventory profile.

Detailed service sizing deferred; infrastructure secondary to fuel integration.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Partial gross/lease stocks, without client handling/inventory profile.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Nigel: decide whether to prioritise investment analysis after the flow case. Manish and Henry: bridge incremental flows to usable stock and service constraints.

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com

